

CSE213 CRA Report

Summer 2026

Topic: Simulating operations of a Private power generation company such as Summit Power		
ID: 2310604	Name: Ridwan Hasan Khandakar	User1: Accountant
		User2: BPDB Officer
ID: 2022752	Name: Kazi Fatema Tuj Johra	User3: HR Manager
		User4: Inventory Manager
ID: 2110304	Name: Rubama Mehnaj	User5: CEO
		User6: Plant Manager
ID: 2431072	Name: Samia Islam	User7: Engineer
		User8: Grid Operator

NOTE:

- Download this template and create a google-docs file with the following **file name format**:
 - MS1_GroupNo_MS1Version1_topicKeyword_id1_id2_id3_id4
 - Ex: MS1_Group90_MS1Version1_IRAS_1234567_2345678_3456789_4567890
- Group Leader will create the google-docs using this template and will share the google-docs file among group members. All of the group members need to work on the same document.
- Group leader must also upload the "viewer share" link of your google-docs file, so that the faculty can view your document anytime and check your progress (link will be given, soon).
- **ONLY ONE SUBMISSION from EACH GROUP**
- **To lock your content, you MUST upload a PDF file. NO LINK will be accepted.**
 - **By 04-June-26**, Saturday, the group leader MUST upload the PDF of your google-docs containing workflows for ALL 64 goals.
 - Among 64 workflows (of the goals) for 8 users, 16 workflows (4 workflows per member, 2 workflows/user), MUST look complete and comprehensive.
 - Submission Link will be given before the deadline.
- **If you fail to submit version-1 of your MileStone-1 by 04-June-26, you will not get any mark for MileStone-1**

Possible event type:

- UIE -- user input to trigger event
- UID -- user input to be considered as data
- OP – display content (output)
- VL – validation check (usually for user input)
- VR – verification check (verify with existing application data)
- DP – fetching data from file system and process it to get some calculated outcome (data processing)

NOTE:

there can be workflows for common **processes** as follows (**BUT, these are NOT GOALS**):

Process-1: Login			
			event type
Workflow:	event-1	Enter username(must be 4 digit number) and password	VL
	event-2	Validate login credentials against database	VL
	event-3	Verify user role and permissions	VR
	event-4	If login is successful, Open dashboard(role)	OP

Process-2: SignOut			
			event type
Workflow:	event-1	Button click for signout	UIE
	event-2	Take user back to login page	UIE

Process-3: SignUp (for example)			
			event type
Workflow:	event-1	Button click for signup	UIE
	event-2	Enter first and last name	UID
	event-3	Create username	UID
	event-4	Create password	UID
	event-5	Re-enter password	VR
	event-6	Validates that passwords match	VR

	event-7	Show success message: “SignUp successful”	OP
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Process-4: Change Password

			event type
Workflow:	event-1	Click “Change Password” button from dashboard	UIE
	event-2	Enter old password	UID
	event-3	Enter new password	UID
	event-4	Re-enter new password for confirmation	UID
	event-5	Validates that new passwords match	VL
	event-6	Verify old password with database	VR
	event-7	If valid, update password in user record	DP
	event-8	Show success message: “Password updated”	OP

Workflow of user specific goals:

User-1 name: Accountant

User-1, Goal-1	Description of Goal-1: Log income from electricity sales		event type
Workflow:	event-1	Login process	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks on “Sales Income” module	UIE
	event-4	System loads sales entry form with fields: Client/Distribution Company Name	OP

		- Electricity units sold (MWh) - Revenue amount - Date -	
	event-5	Accountant fills in sales information	UID
	event-6	Accountant selects transaction date	UID
	event-7	Accountant clicks “Log Sale” button	UIE
	event-8	System validates all fields and checks numeric format	VL
	event-9	If valid, system stores income record	DP
	event-10	Confirmation message: “Electricity sale logged successfully”	OP
	event-11	Entry appears in sales history table	Dp
User-1, Goal-2	Description of Goal-2: Track fuel and operational expenses		
Workflow:	event-1	login process	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Expenses” module	UIE
	event-4	System loads expense form with fields: - Expense category (Fuel / Maintenance / Utility / Other) - Amount - Date	OP
	event-5	Accountant selects expense type	UID
	event-6	Accountant enters amount and selects date	UID
	event-7	Accountants clicks “Add Expense”	UIE
	event-8	System validates input values	VL

	event-9	System saves expense record	DP
	event-10	Confirmation message : “Expense logged successfully”	OP
	event-11	Expense entry appears in monthly summary table	DP
User-1, Goal-3	Description of Goal-3: Manage salary payments		
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Salary Management”	UIE
	event-4	System displays employees salary table: - Employee Name - ID - Salary - Payment Status	DP
	event-5	Accountant selects month from dropdown	UID
	event-6	System filters salary records for selected month	DP
	event-7	Accountant selects unpaid salary entries	UIE
	event-8	Accountant clicks “Process Payment”	UIE
	event-9	System verifies selected records	VR
	event-10	System updates salary payments status	DP
	event-11	Confirmation message : “Salaries processed successfully”	OP
User-1, Goal-4	Description of Goal-4: Generate monthly profit/loss reports		
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Reports” module	UIE
	event-4	System displays month/year selection dropdown	OP

	event-5	Accountant selects reporting period	UID
	event-6	System validates selected period	VL
	event-7	System retrieves: - Sales income - Expenses - Salary costs - Fuel costs	DP
	event-8	System calculates profit/loss	DP
	event-9	Profit/Loss summary displayed with breakdown	OP
User-1, Goal-5	Description of Goal-5: Monitor fuel inventory valuation		
Workflow:	event-1	Login Process	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Fuel Inventory Valuation” module	UIE
	event-4	System displays inventory table: - Fuel type - Quantity - Unit cost	DP
	event-5	Accountant selects category/date range	UID
	event-6	System filters inventory records	DP
	event-7	System calculates total value: Quantity x Unit cost	DP
	event-8	Total valuation displayed	OP
User-1, Goal-6	Description of Goal-6: Approve high-cost purchase requests		
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Purchase Approvals” module	UIE

	event-4	System loads pending purchase request list: • Cost • Description • Department	DP
	event-5	Accountant reviews requests	OP
	event-6	Accountant selects request	UID
	event-7	Accountant clicks “Approve” or “Reject”	UIE
	event-8	System verifies amount threshold	VR
	event-9	System updates request status	DP
	event-10	Confirmation message shown	OP
	event-11	Purchase history table updated	DP
User-1, Goal-7	Description of Goal-7: Handle tax and audit records		
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load Accountant dashboard	OP
	event-3	Accountant clicks “Tax & Audit Records” module	UIE
	event-4	System displays categorized records: • Tax Returns • VAT • Audit Logs	DP
	event-5	Accountant selects record type/year	UID
	event-6	System filters records	DP
	event-7	Accountant selects a record	UIE
	event-8	System displays record details	OP
User-1, Goal-8	Description of Goal-8: Generate financial statements		
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load Accountant dashboard	OP

	event-3	Accountant clicks “Financial Statements” module	UIE
	event-4	System displays statement type dropdown: • Balance Sheet • Income Statement • Cash Flow Statement	OP
	event-5	Accountant selects statement type and reporting period	UID
	event-6	System validation selection	VL
	event-7	System retrieves: • Income records • Expense records • Tax data • Asset data	DP
	event-8	System generates formatted financial statement	DP
	event-9	Statement preview displayed	OP

User-2 name: BPDB Officer			
User-2, Goal-1	Description of Goal-1: Submit Power Demand Request		event type
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load BPDB Officer dashboard	OP
	event-3	Click “Power Demand Request” module	UIE
	event-4	System loads demand request form	OP
	event-5	Enter required power capacity (MW)	UID
	event-6	Select supply period from dropdown	UID
	event-7	Enter additional remarks	UID
	event-8	Click “Submit Request” button	UIE
	event-9	System validates entered information	VL
	event-10	System stores request in database/file	DP
	event-11	Confirmation message displayed	OP
User-2, Goal-2	Description of Goal-2: View Power Supply Agreements		
Workflow:	event-1	Login process execution	See login process
	event-2	Load BPDB dashboard	OP
	event-3	Click "Supply Agreements" module	UIE
	event-4	System retrieves agreement records	DP

	event-5	Display agreements in TableView	OP
	event-6	Select displays agreement record	UIE
	event-7	System displays agreement details	OP
	event-8	Review agreement information	DP
	event-9	Click “Download Agreement” button	UIE
	event-10	System generates downloadable file	DP
User-2, Goal-3	Description of Goal-3: Monitor Daily Power Supply		
Workflow:	event-1	Login process execution	See login process
	event-2	Load dashboard	OP
	event-3	Open "Power Supply Monitoring" module	UIE
	event-4	System retrieves daily supply records	DP
	event-5	Displays supplied power data in table	OP
	event-6	Select a date range	UID
	event-7	System filtered records	DP
	event-8	Calculated total supplied power	DP
	event-9	Displays summary information	OP
	event-10	Save monitoring act	UID
User-2, Goal-4	Description of Goal-4: View Invoices and Payment Records		
Workflow:	event-1	Login process execution	See login process
	event-2	Load dashboard	OP
	event-3	Click “Invoice & Payments” module	UIE

	event-4	Retrieve invoice records	DP
	event-5	Displays invoice records	DP
	event-6	Selects invoice record	UIE
	event-7	View payment details	OP
	event-8	Check payment details	DP
	event-9	Click “Download Invoice” button	UIE
	event-10	System generates invoice file	DP
User-2, Goal-5	Description of Goal-5: Submit Power Supply Complaint		
Workflow:	event-1	Events of login process to be executed	See login process
	event-2	Load dashboard	OP
	event-3	Click “Submit Complaint” module	UIE
	event-4	System loads complaint form	OP
	event-5	Select complaint category	UID
	event-6	Enter complaint details	UID
	event-7	Select affected supply date	UID
	event-8	Click “Submit” button	UIE
	event-9	System validates information	VL
	event-10	Complaint stored in records	DP
	event-11	Confirmation displayed	OP
User-2, Goal-6	Description of Goal-6: Request Additional Power Supply		
Workflow:	event-1	Events of login process to be executed	See login

			process
	event-2	Load dashboard	OP
	event-3	Open “Additional supply Request” module	UIE
	event-4	Enter required additional capacity	UID
	event-5	Select requested date range	UID
	event-6	Enter request justification	UID
	event-7	Clicks “Submit Request”	UIE
	event-8	Validate request information	VL
	event-9	Store request record	DP
	event-10	Notify Plant Manager	OP
User-2, Goal-7	Description of Goal-7: View Generation and Supply Reports		
Workflow:	event-1	Events of login process to be executed	See login process
	event-2	Load dashboard	OP
	event-3	Click “Reports” module	UIE
	event-4	Select report type	UID
	event-5	Select reporting period	UID
	event-6	System retrieves report data	DP
	event-7	Generate report summary	DP
	event-8	Displays report in table/chart	OP
	event-9	Click “Export Report”	UIE
	event-10	Download report file	DP
User-2, Goal-8	Description of Goal-8: Review Contract Performance		
Workflow:	event-1	Events of login process to be executed	See login

			process
	event-2	Load dashboard	OP
	event-3	Open “Contract Performance” module	UIE
	event-4	Retrieves contract records	DP
	event-5	Displays contract list	OP
	event-6	Select contract	UIE
	event-7	System calculates fulfillment statistics	DP
	event-8	Displays performance indicators	OP
	event-9	Generate performance summary	DP
	event-10	Export summary report	UIE

User-3 name: **HR Manager**

User-3 name: HR Manager			
User-3, Goal-1	Description of Goal-1: Add New Employee		event type
Workflow:	event-1	Events of login process to be executed	See login process
	event-2	If login is successful, load "HR Manager" dashboard	OP
	event-3	Select "Add New Employee" option	UIE
	event-4	Fillup employee details: - Employee ID - Name - Date of Birth - Email - Position - Department - Salary	UID
	event-5	Verify if the provided information or data is valid	VL
	event-6	Save employee information	DP
	event-7	Show "Employee added successfully ! " notification	OP
User-3, Goal-2	Description of Goal-2: Employee Award Management		
Workflow:	event-1	Event of login process to be executed See login process	See login process
	event-2	If login is successful, load "HR Manager" dashboard	OP

	event-3	Select “Award Management” option	UIE
	event-4	Fill up details: - Employee ID - Employee Name - Award Name - Award Type - Award Date	UID
	event-6	Selects “Save” button	UIE
	event-7	Verify if given data is valid	DP
	event-8	Saves Employee Award information	VL
	event-9	Show notification “Employee Award information saved successfully !” notification	OP
User-3, Goal-3	Description of Goal-3: Employee Performance Grading		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “HR Manager” dashboard	OP
	event-3	Select “Performance Grading” option	UIE
	event-4	Fill up details- - Employee ID - Attendance score - Teamwork score - Efficiency score - Evaluation start month - Evaluation end month - Evaluation year - Final Rating	UID
	event-5	Click “Save” button	UIE
	event-6	Verify if provided data is valid	VL
	event-7	Calculates final rating	
	event-8	Save performance grading information	DP
	event-9	Show “Performance grading saved successfully !” notification	OP

User-3, Goal-4	Description of Goal-4: Employee Medical Clearance		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “HR Manager” dashboard	OP
	event-3	Select “Employee Medical Clearance ” option	UIE
	event-4	Fillup details: - Clearance ID - Employee ID - Employee Name - Doctor Name - Medical status - Checkup Date	UID
	event-5	Verify if given data are valid	VL
	event-6	Click “Save” button	UIE
	event-7	Saves employee medical clearance information	DP
	event-8	Show “Employee medical clearance recorded successfully !” notification	OP
User-3, Goal-5	Description of Goal-5: Employee Training Management		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “HR Manager” dashboard	OP
	event-3	Select “Manage Employee Training Program” option	UIE
	event -4	Fill up details: - Select or enter employee ID - Select or enter employee Name - Select training - Select trainer name - Select status	
	event-5	Clicks “Save” button	UIE
	event-6	Verify if given data are valid	VL
	event-7	Save training record and display on table	DP, OP

User-3, Goal-6	Description of Goal-6: Create Notice		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “HR Manager” dashboard	OP
	event-3	Select “Create Notice” option	UIE
	event-4	Enter notice details: - Notice Id - Created by - Expiry date - Publish Date - Subject - Content	UID
	event-5	Click “Create” button	UIE
	event-6	Verify if given data are valid	VL
	event-7	Creates and saves notice	DP
	event-8	Show “Notice created” alert	OP
	event-9	Selects “View” button	UIE
	event-10	Shows notice details on a new window	OP
User-3, Goal-7	Description of Goal-7: View Employee Details		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “HR Manager” dashboard	OP
	event-3	Select “View Employee details” option	UIE
	event-4	Fill the details- - Enter department	UID
	event-5	Click “Load” button	UIE
	event-6	Show employee list on table with employee details	OP
User-3,	Description of Goal-8: Employee Leave Management		

Goal-8			
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “HR Manager” dashboard	OP
	event-3	Select “Manage Employee Leave” option	UIE
	event-4	Fillup details: - Leave ID - Employee ID - Leave Type - Leave Reason - Total days - Start date - End date	UID
	event-5	Clicks “Save” button	UIE
	event-6	Verify if the provided data are valid	VL
	event-7	Save employee leave information	DP
	event-8	Show “Employee leave information recorded successfully !” notification	OP

User-4 name: Inventory Manager			
User-4, Goal-1	Description of Goal-1: Add Equipment		event type
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “Inventory Manager” dashboard	OP
	event-3	Select “Add Equipment” option	UIE
	event-4	Fillup equipment details: - ID - Name - Category - Quantity - Vendor - Purchase Date	UID
	event-5	Clicks “Save” button	UIE

	event-6	Verify if the provided product data is valid	VL
	event-7	Equipment created and saved	DP, OP
	event-8	Show "Equipment added successfully !" notification	OP
	event-9	Clicks "View" button	UIE
	event-10	Shows equipment list on another window	OP
User-4, Goal-2	Description of Goal-2: Equipment Maintenance History		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "Inventory Manager" dashboard	OP
	event-3	Select "Equipment Maintenance History" option	UIE
	event-4	Fill up details- - Enter employee ID - Maintenance ID - Spare Parts used - Quantity used - Technician - Maintenance Date	UID
	event-5	Clicks "Save" button	UIE
	event-6	Verify if given data are valid	VL
	event-7	Save equipment maintenance history	DP
	event-8	Shows "Equipment maintenance history added successfully!" notification	OP
User-4, Goal-3	Description of Goal-3: Parts Issuance		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "Inventory Manager" dashboard	OP
	event-3	Select "Parts Issuance" option	UIE

	event-4	Fill up issuance details: - Select equipment ID - Requester employee ID - Select quantity required - Issue Date	UID
	event-5	Select "Issue Parts" button	UIE
	event-6	Verify if given data are valid	VL
	event-6	Save parts issuance information	DP
	event-7	Show "Parts Issuance created successfully!"	OP
	event-8	Click "Load table view" button	UIE
	event-9	Read all parts issuance records from bin file	DP
	event-10	Display all parts issuance records on the table	OP
User-4, Goal-4	Description of Goal-4: Offer Purchase Tender		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "Inventory Manager" dashboard	OP
	event-3	Select "Offer Tender" option	UIE
	event-4	Enter tender details: - Tender ID - Tender title - Equipment Name - Closing Date	UID
	event-5	Clicks "Create Tender" button	UIE
	event-6	Verify if given data are valid	VL
	event-7	Created and saved tender information	DP
	event-8	Show "Tender Added Successfully !" notification	OP
User-4, Goal-5	Description of Goal-5: Vendor Management		
Workflow:	event-1	events of login process to be executed	See login process

	event-2	If login is successful, load “Inventory Manager” dashboard	OP
	event-3	Select “Vendor Management” option	UIE
	event-4	Fill up details: - Name - ID - Equipments Supplied - Email	UID
	event-5	Clicks “Save” button	UIE
	event-6	Verify if provided data is valid	VL
	event-7	Saves Vendor information	DP
	event-8	Show “Vendor added successfully !” notification	OP
	event-9	Click “Load table view” button	UIE
	event-10	Load and display vendor information from the vendor list onto the table	DP, OP
User-4, Goal-6	Description of Goal-6: Inventory Report		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load “Inventory Manager” dashboard	OP
	event-3	Select “Inventory Report” option	UIE
	event-4	Select criteria for report generation: - Report ID - Report type - Inventory category	UID

		- Report Date - Item - quantity	
	event-5	Select "Save" button	UIE
	event-6	Verify if the given data is valid	VL
	event-7	Create and save inventory report	DP
	event-8	Show "Inventory report created successfully !" notification	OP
User-4, Goal-7	Description of Goal-7: Fuel Consumption Tracking		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "Inventory Manager" dashboard	OP
	event-3	Select "Fuel Consumption Tracking" option	UIE
	event-4	Fillup equipment details: - Consumption ID - Quantity used - Fuel Type - Plant Unit - Consumption Date	UID
	event-5	Clicks "Save" button	VL
	event-6	Verify if the given data is valid	OP
	event-7	Save fuel consumption information	DP
	event-8	Show "Fuel consumption recorded successfully !" notification	OP
User-4, Goal-8	Description of Goal-8: View Inventory Report		
Workflow:	event-1	events of login process to be executed	See login process
	event-2	If login is successful, load "Inventory Manager" dashboard	OP

	event-3	Select “View Inventory Report” option	UIE
	event-4	Select report type	UID
	event-5	Click “Load Table View” button	UIE
	event-6	Filter Inventory reports according to report type	VR
	event-7	Display filtered inventory reports	OP

User-5 name: CEO			
User-5, Goal-1	Description of Goal-1: View plant performance summary		event type
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "Plant Performance" menu.	UIE
	event-4	The system fetches a “View plant performance summary” view.	DP
	event-5	The CEO chooses the starting and to ending date from the date picker.	UID
	event-6	Admin selects date range (e.g., last 7 days) to compare output.	UID
	event-7	The system fetches historical generation data such as starting- ending date, total generation, total power loss and performance status and displays a TableView.	DP, OP
User-5, Goal-2	Description of Goal-2: Check financial summary (profit/loss)		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN

			PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "Financial Summary" button.	UIE
	event-4	The CEO chooses the from and to date joining date from the date picker.	UID
	event-5	The CEO clicks the "Load Financial Summary" button.	UIE
	event-6	The system shows total revenue, total fuel cost, and operational cost through a TableView.	OP
User-5, Goal-3	Description of Goal-3: Set or change tariff rates		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the " Set or change tariff rates" button.	UIE
	event-4	Admin selects a customer type from the ComboBox	UID
	event-5	Admin enters a new rate in the TextField	VL
	event-6	Admin selects a set tariff date from the Datepicker	UID
	event-7	Admin clicks the "Set Tariff" button to submit the change.	UIE
	event-8	The system shows "Tariff updated successfully".	OP
	event-9	Updated details are shown in the TableView	OP
User-5, Goal-4	Description of Goal-4: Approve fuel purchase request		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP

	event-3	The CEO clicks the "Approve fuel purchase request" button.	UIE
	event-4	The CEO clicks the "pending fuel request" button.	UIE
	event-5	The system shows "TableView of requests and approval status".	OP
	event-6	The CEO clicks the "approve" button.	UIE
	event-7	The CEO clicks the "approve not" button If not needed to approve.	UIE
User-5, Goal-5	Description of Goal-5: Add or remove an employee		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "Add or remove an employee " button.	UIE
	event-4	CEO enters employee name, role, and employee ID.	UID
	event-5	The system checks that the employee ID is not already used.	VR
	event-6	The system shows "Employee added successfully" through an Alert.	OP
	event-7	The system shows "Employee removed successfully" through an Alert.	OP
User-5, Goal-6	Description of Goal-6: View total registered customers		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "View Total Registered Customers" button.	UIE

	event-4	The CEO chooses the customer joining date from the date picker.	UID
	event-5	The system fetches a total number of registered customers in a TableView containing TableColumns.	DP
	event-6	The system shows the count of industrial, commercial, and residential customers in TableColumns.	OP
User-5, Goal-7	Description of Goal-7: View customer complaint summary		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "Complaint Summary" button.	UIE
	event-4	Selecting the combo box selection and selecting date from the Date Picker.	DP
	event-5	After clicking the "View Complaints" button it fetches the total complaints (resolved/pending) from customer records.	UIE
	event-6	The system shows the number of complaints today, this week, and still pending in a TableView with Complaint ID, Date, Status and Category.	OP
User-5, Goal-8	Description of Goal-8: Generate weekly report		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load CEO dashboard	OP
	event-3	The CEO clicks the "Generate Weekly Report" button.	UIE
	event-4	The CEO chooses the starting and ending date from the date picker.	UID

	event-5	The CEO clicks the "Show Performance" button.	UIE
	event-6	The TableView displays the date ,total generation, total power loss and performance status on screen.	OP
	event-7	The CEO clicks the "Generate Weekly Report" button.	UIE
	event-8	Shows the filtered weekly report.	OP

User-6 name: Plant Manager

User-6, Goal-1	Description of Goal-1: Monitor current power generation		event type
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Plant Manager dashboard	OP
	event-3	Plant Manager clicks the "Power Generation History" button.	UIE
	event-4	Admin selects a Unit Name from the ComboBox	UID
	event-5	Admin selects a Unit No from the ComboBox	UID
	event-6	Selecting date from the Date Picker.	DP
	event-7	Plant Manager clicks the "View Generation" button.	UIE
	event-8	The TableView displays the date ,unit name, unit no and status on screen.	OP
User-6, Goal-2	Description of Goal-2: Check fuel level		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Plant Manager dashboard	OP
	event-3	Plant Manager clicks the "Fuel Level History" button.	UIE
	event-4	Admin selects a Unit Name from the ComboBox	UID
	event-5	Admin selects a Unit No from the ComboBox	UID
	event-6	Selecting date from the Date Picker.	DP
	event-7	Plant Manager clicks the "Check Fuel Levels" button.	UIE

	event-8	The TableView displays the date ,unit name, fuel type and status on screen.	OP
User-6, Goal-3	Description of Goal-3: Schedule maintenance for a unit		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Plant Manager dashboard	OP
	event-3	The Plant Manager clicks the "Maintenance Schedule" button.	UIE
	event-4	Admin selects a Unit Name from the ComboBox	UID
	event-5	Admin selects a Unit No from the ComboBox	UID
	event-6	Selecting date from the Date Picker.	DP
	event-7	Plant Manager clicks the "View Schedule" button.	UIE
	event-8	The TableView displays the date ,unit name, unit no and status on screen.	OP
User-6, Goal-4	Description of Goal-4: Start or stop a generating unit		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Plant Manager dashboard	OP
	event-3	The Plant Manager clicks the "Generation Units" button.	UIE
	event-4	Admin selects a Unit Name from the ComboBox	UID
	event-5	Admin selects a Unit No from the ComboBox	UID
	event-6	Selecting date from the Date Picker.	DP
	event-7	The Plant Manager clicks the "Maintenance History" button.	UIE

	event-8	The TableView displays the date ,unit name, unit no and status on screen.	OP
User-6, Goal-5	Description of Goal-5: View maintenance history		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Plant Manager dashboard	OP
	event-3	The Plant Manager clicks the "Maintenance History" button.	UIE
	event-4	Admin selects a Unit Name from the ComboBox	UID
	event-5	Admin selects a Unit No from the ComboBox	UID
	event-6	Selecting date from the Date Picker.	DP
	event-7	The Plant Manager clicks the "Maintenance History" button.	UIE
	event-8	The TableView displays the date ,unit name, unit no and status on screen.	OP
User-6, Goal-6	Description of Goal-6: Generate daily shift report		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Plant Manager dashboard	OP
	event-3	Plant Manager clicks "Shift Report".	UIE
	event-4	Admin selects a Unit Name from the ComboBox	UID
	event-5	Admin selects a Unit No from the ComboBox	UID
	event-6	Selecting date from the Date Picker.	DP
	event-7	The Plant Manager clicks the "View Report" button.	UIE

	event-8	The TableView displays the date ,unit name, unit no and status on screen.	OP
User-6, Goal-7	Description of Goal-7: Raise fuel request to Admin		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	If login is successful, load Plant Manager dashboard	OP
	event-3	Plant Manager clicks the "Request Fuel" button.	UIE
	event-4	Admin selects a Unit Name from the ComboBox	UID
	event-5	Admin selects a Fuel Type from the ComboBox	UID
	event-6	Selecting date from the Date Picker.	DP
	event-7	The Plant Manager clicks the "Request Fuel" button.	UIE
	event-8	The TableView displays the date ,unit name, fuel type and status on screen.	OP
User-6, Goal-8	Description of Goal-8: Log an unplanned breakdown		
<u>Workflow:</u>	event-1	Login process	CHECK LOGIN PROCESS
	event-2	Plant Manager clicks the "Log Breakdown" button.	UIE
	event-3	Admin selects a Unit Name from the ComboBox	UID
	event-4	Admin selects a Unit No from the ComboBox	UID
	event-5	Selecting date from the Date Picker.	DP
	event-6	The Plant Manager clicks the "View Breakdown Report" button.	UIE
	event-7	The TableView displays the date ,unit name, unit no and status on screen.	OP

User-7 name: Engineer

User-7, Goal-1	Description of Goal-2: Monitor Generator Status		event type
Workflow:	event-1	Login process execution	See login process
	event-2	If login is successful, load the Engineer Dashboard where the engineer can access all engineering functions.	OP

	event-3	Engineer clicks on "Monitor Generator Status" module.	UIE
	event-4	System loads Generator Status form with fields: • Generator ID • Plant Unit	OP
	event-5	Engineer enters Generator ID, Engineer selects Plant Unit, Engineer clicks "Search" button.	UID,UIE
	event-6	If a matching generator is found, display Running Status, Power Output, Voltage, Frequency, Fuel Level, Engine Temperature, Running Hours, and Last Maintenance Date. Otherwise display "Generator not found."	DP,OP
User-7, Goal-2	Description of Goal-3: Schedule Preventive Maintenance		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Engineer Dashboard.	OP
	event-3	Select Preventive Maintenance option	UIE
	event-4	Enter Maintenance ID, Generator ID, Plant Unit, Maintenance Type, Scheduled Date, Engineer Name, Priority, Maintenance Status and Remarks.	UID
	event-5	The system validates the inputs and checks that the scheduled date is not in the past.	VL,VR
	event-6	Save the maintenance schedule, refresh the table and display a success message.	DP,OP
User-7, Goal-3	Description of Goal-4: Report Equipment Faults		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Engineer Dashboard.	OP
	event-3	Select Report Equipment Faults option.	UIE
	event-4	Enter Fault ID, Equipment ID, Generator ID, Fault Category, Fault Severity, Report Date and Fault Description.	UID

	event-5	The system validates all fields and ensures the report date is not in the future.	VL,VR
	event-6	Save the fault report, refresh the table and display a success message-.	DP,OP
User-7, Goal-4	Description of Goal-5: Update Maintenance Status		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Engineer Dashboard.	OP
	event-3	Select Update Maintenance Status option.	UIE
	event-4	Enter Maintenance ID, Generator ID, Maintenance Date, Maintenance Status, Updated By and Remarks.	UID
	event-5	Validate the inputs and verify that the Maintenance ID exists in the database before updating the maintenance status.	VL,VR
	event-6	Save the updated maintenance status, refresh the table and display a success message.	DP,OP
User-7, Goal-5	Description of Goal-6: Analyze Equipment Efficiency		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Engineer Dashboard.	OP
	event-3	Select Equipment Efficiency Button.	UIE
	event-4	Enter Equipment ID, Generator ID, Power Output, Fuel Consumption and Analysis Date. Click Calculate.	UID
	event-5	The system validates all inputs and calculates Equipment Efficiency = Power Output / Fuel Consumption.	VL,VR
	event-6	Display the calculated efficiency. When Save is clicked, store the record and refresh the table.	DP,OP

User-7, Goal-6	Description of Goal-7: Monitor Fuel Consumption			
Workflow:	event-1	Login process execution.		See login process
	event-2	Open Engineer Dashboard .		OP
	event-3	Select Monitor Fuel Consumption.		UIE
	event-4	Enter Generator ID, Fuel Type, Fuel Consumed, Operating Hours, Monitoring Date and Remarks.		UID
	event-5	The system validates all inputs, ensuring numeric values are positive and the monitoring date is not in the future.		VL,VR
	event-6	Save the fuel consumption record, refresh the table, and display a success message .		DP,OP
User-7, Goal-7	Description of Goal-8: Manage Spare Parts Requirements			
Workflow:	event-1	Events of login process to be executed. See Login Process.		See login process
	event-2	If login is successful, load Engineer Dashboard.		OP
	event-3	Select Manage Spare Parts Requirement.		UIE
	event-4	Enter Request ID, Part Name, Quantity, Supplier Name, Request Date and Request Status.		UID
	event-5	Validate all inputs, ensuring quantity is positive and request date is today or later.		VL,VR
	event-6	Save the spare parts request, refresh the table and display a success message.		DP,OP
User-7, Goal-8	Description of Goal-9: Generate Engineering Performance Reports			
Workflow:	event-1	Events of login process to be executed. See Login Process		See login process
	event-2	If login is successful, load Engineer Dashboard		OP
	event-3	Select Engineering Performance Report.		UIE
	event-4	Select Generator ID, Report Date and click		UID

		Generate Report.	
	event-5	The system validates the inputs, calculates Efficiency Percentage, and determines the Performance Status (Excellent, Good, Average or Poor).	VL,VR
	event-6	Save the engineering performance report, refresh the table, and display a success message.	DP,OP
User-7, Goal-9	Description of Goal-10: Maintenance Performance Analytics		
Workflow:	event-1	Login process execution.	See Login Process
	event-2	Open Engineer Dashboard.	OP
	event-3	Select Maintenance Performance Analytics.	UIE
	event-4	View engineering performance records and optionally select a Performance Status filter.	UID
	event-5	The system filters records according to the selected status and prepares performance statistics.	DP
	event-6	Display the filtered table and update the performance pie chart showing Excellent, Good, Average and Poor categories.	OP

User-8 name: Grid Operator			
User-8, Goal-1	Description of Goal-1: Monitor Grid Status		event type
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP

	event-3	Select Monitor Grid Status option.	UIE
	event-4	Enter Grid ID and click Search.	UID
	event-5	The system searches for the Grid ID and verifies whether the record exists.	VL,VR
	event-6	If found, display Grid Section, Grid Status, Voltage, Frequency and Monitoring Date. Otherwise display "No record found with this Grid ID."	DP,OP
User-8, Goal-2	Description of Goal-2: Monitor Load Demand		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select Monitor Load Demand option.	UIE
	event-4	Enter Demand ID and click Search.	UID, UIE
	event-5	The system verifies whether the entered Demand ID exists and calculates Available Capacity and Demand Status.	VL,VR
	event-6	Display Grid Section, Current Load, Available Capacity, Demand Status and Monitoring Date. Otherwise display "No record found with this Demand ID."	DP,OP
User-8, Goal-3	Description of Goal-3: Synchronize Generator with National Grid		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP

	event-3	Select Synchronize Generator with National Grid option.	UIE
	event-4	Enter Synchronization ID, Generator ID, Grid Section, Generator Voltage, Grid Voltage, Synchronization Status and Synchronization Date.	UID
	event-5	The system validates all inputs and verifies that Generator Voltage equals Grid Voltage and the date is valid.	VL,VR
	event-6	Save the synchronization record, clear the form and display "Synchronization record saved successfully."	DP,OP
User-8, Goal-4	Description of Goal-4: Respond to Grid Fault		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select Respond to Grid Fault option.	UID
	event-4	Enter Fault ID, Grid Section, Fault Type, Fault Date, Response Action and Fault Status.	VL
	event-5	The system validates all inputs, ensures the fault date is not in the future and checks response length.	VR
	event-6	Save the fault response, clear the form and display "Grid fault response saved successfully."	DP
User-8, Goal-5	Description of Goal-5: Schedule Load Distribution		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process

	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select Load Distribution option.	UIE
	event-4	Enter Distribution ID, Grid Section, Load (MW), Distribution Status, Distribution Date and Remarks.	UID
	event-5	The system validates all inputs, verifies Load is greater than zero and Distribution Date is not in the past.	VL,VR
	event-6	Save the load distribution record, clear the form and display "Load Distribution saved successfully."	DP,OP
User-8, Goal-6	Description of Goal-6: View Grid Performance Report		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select View Grid Performance Report option.	UIE
	event-4	Enter Report ID, Grid Section, Total Load, Available Capacity and Report Date, then click Generate Report.	UID
	event-5	The system validates the inputs, calculates Grid Efficiency and determines Performance Status (Excellent, Good, Average or Poor).	VL,VR
	event-6	Save the report, refresh the table and display "Grid Performance Report saved successfully."	DP,OP
User-8, Goal-7	Description of Goal-7: Search Transmission Line Information		

Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select Transmission Line Information option.	UIE
	event-4	Enter Line ID, Line Name, Source Station, Destination Station, Voltage Level, Line Length and Line Status.	UID
	event-5	The system validates all inputs and verifies that Voltage Level and Line Length are greater than zero.	VL,VR
	event-6	Save the transmission line information, refresh the table and display "Transmission Line saved successfully."	DP,OP
User-8, Goal-8	Description of Goal-8: Record Grid Operation Log		
Workflow:	event-1	Events of login process to be executed. See Login Process.	See login process
	event-2	If login is successful, load Grid Operator Dashboard.	OP
	event-3	Select Record Grid Operation Log option.	UIE
	event-4	Enter Operation ID, Grid ID, Grid Section, Operation Type, Grid Status, Voltage, Frequency, Operation Date, Operator Name and Remarks.	UID
	event-5	The system validates all inputs, verifies numeric values and ensures the operation date is not in the past.	VL,VR
	event-6	Save the operation log, refresh the table and display "Operation Log saved successfully."	DP,OP

